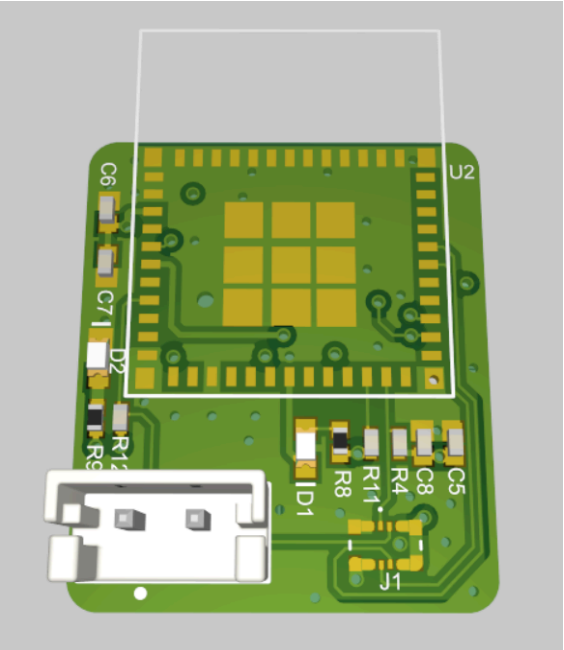
IMU Golf Ball PCB

1. Overview

Designed a compact 4-layer PCB in Altium to fit inside a regulation golf ball and support IMU-based motion tracking. The ESP32 leverages FreeRTOS to run deterministic IMU sampling and sensor fusion tasks in parallel with communication and logging processes.

Primary focus:

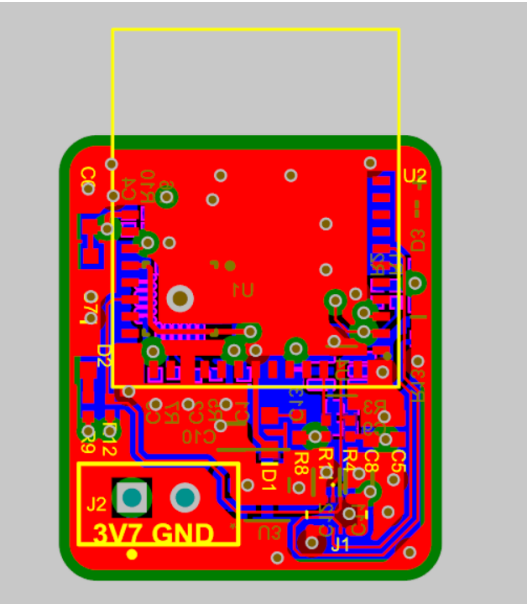
- Extreme miniaturization
- Signal integrity
- Power integrity
- Mechanical robustness under high acceleration



2. Board Architecture

Layer Stack (4-Layer):

- Top: Signals
- Inner 1: Dedicated GND plane
- Inner 2: Dedicated 3.3V Power plane
- Bottom: Signals

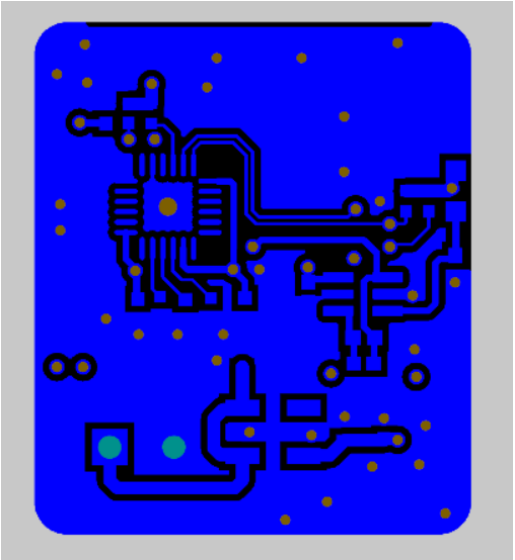
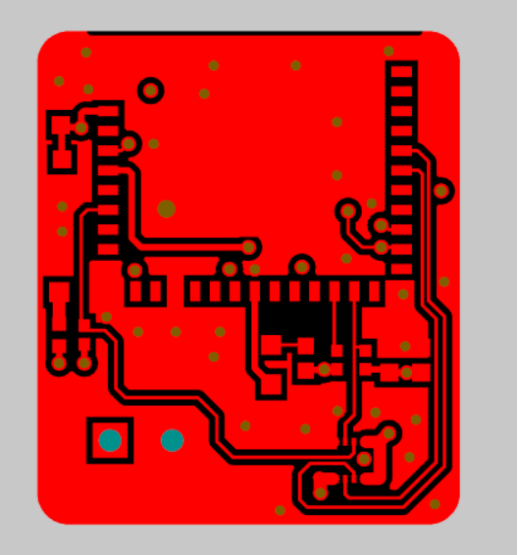


3. Routing Strategy

- Short sensor communication traces
- Wide power traces
- Minimal via transitions
- Continuous ground reference under signal layers

Controlled return paths enabled by solid ground plane.

4. Key Design Considerations

- 4-layer stack for improved signal integrity
- Dedicated ground plane for stable return paths

- Dedicated power plane for low-noise supply
- High-density routing under strict space constraints
- Designed for high-acceleration environment